REVIEW PROBLEMS - ANSWERS

QUESTION 1

a.
We are 95% confidence that the long-run proportion of spins that the racquet would land up is between 0.322 and 0.598 for racquets similar to the one in the study

b.
The confidence interval should get narrower (less wide) with the larger sample size, but the midpoint would still be the same.

QUESTION 2

They will have the same margin-of-error

QUESTION 3

a.
Calories

b.
Poultry

c.
113

d.
42

e.
645

QUESTION 3

a.
One observational unit is one adult American.

b.
The variable is the number of close friends that the adult American has. This variable is quantitative.

c.
Answer: C

d.
"If you took another sample of 1467 adult Americans, there is a 95% chance that its sample mean would fall within this interval" is incorrect because we are not trying to capture sample means – we are trying to capture the population mean.

QUESTION 4

a)

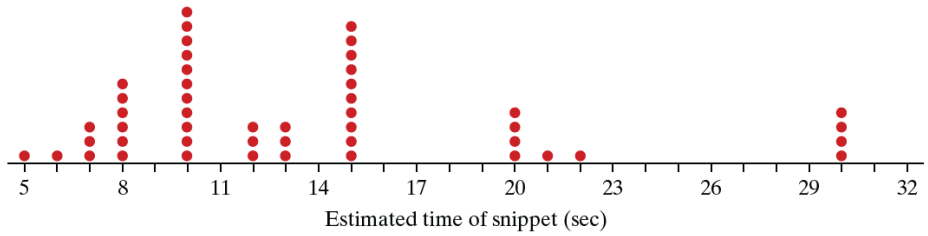
On average, can people accurately estimate the length of a short song snippet?

b)

H_0: $\mu = 10$ seconds

H_a: $\mu \neq 10$ seconds

Here is the dotplot of this study showing the results (in seconds) for 48 students trying to estimate the length of a 10-second song snippet. The sample mean is 13.71 seconds, the median is 10 seconds, and the sample standard deviation is 6.50 seconds.



c)

In this sample of 48 students, the distribution of estimated times of the snippet ranges from 5 seconds to 30 seconds, is centered around 10 seconds, with "spikes" at 10, 15, 20, and 30 seconds. The distribution is somewhat skewed to the right with standard deviation of 6.5 second representing the typical distance from the individual values to the mean.

d)

Yes, it is possible to observe a sample mean length (in a sample of size 48) of 13.71 seconds if the true average length of a song is 10 seconds. However, a single sample cannot tell us whether this is likely/plausible or not.

e)

Because we have a sample size of 48 and the sample distribution is not terribly skewed, we could consider using the theory-based approach.

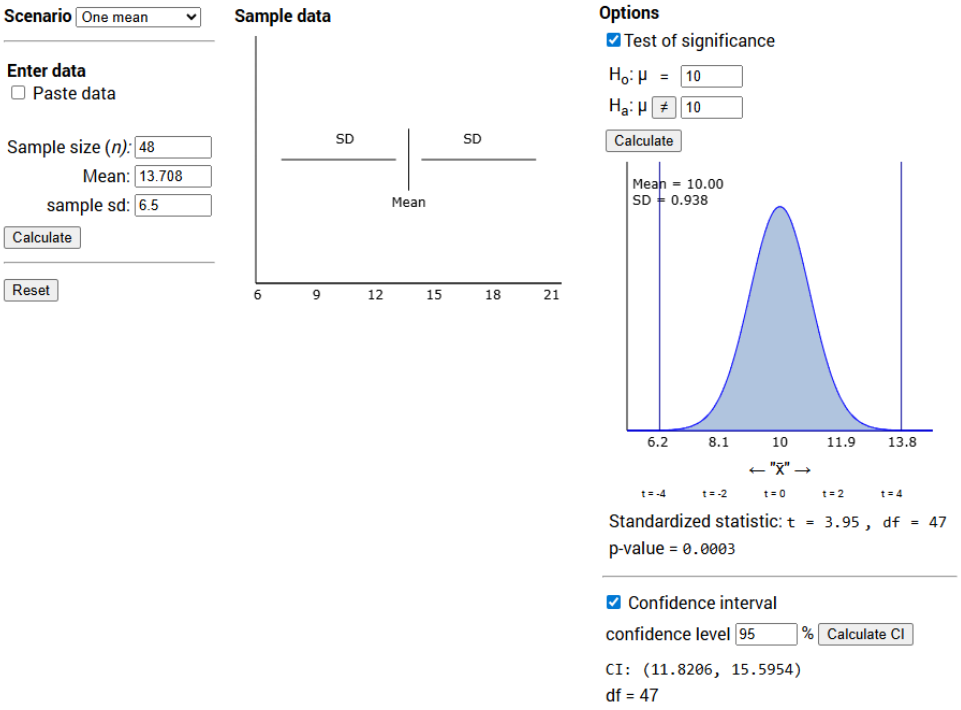
f)

$$standardized\ statistic = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$t = (13.71 - 10) / (6.50 \div \sqrt{(48)}) \approx 3.95$

g)

Yes, the analysis is correct. The sample size of 48, sample mean of 13.708 and sample standard deviation of 6.50 are correct. The hypothesized value of 10 is also correct and a two-sided test was conducted correctly.



h)
The p-value of 0.0003 is the probability of observing a sample mean time estimate of 13.708s or more extreme values on both tails, if the population mean time estimate is really 10s (null hypothesis is true).

i)
The population mean

j)
This study was conducted on a convenience sample of 48 college students at a US university. First, we need to recognize that this was not a random sample. However, even though this was not a random sample, it is doubtful that there was something substantially different or unique about this group of 48 students in their time estimation ability than those of the entire student body at the college. In fact, with regards to estimating elapsed time, it might be reasonable to consider this sample as representative of all college students in the United States, even though the students were selected from just one US college.

k)
We have very strong evidence (p-value = .0003 is small) that population mean time estimate differs from 10 seconds for college students in the United States.

QUESTION 5

a.
Age (quantitative variable) – comparison of means
Weight (quantitative variable) – comparison of means
Gender (categorical variable) – comparison of proportions
Number of cigarettes smoked (quantitative variable) – comparison of means

Whether the person made a previous attempt to quit smoking (categorical variable) – comparison of proportions

b.
The researchers would hope to fail to reject the null hypotheses in these tests because the null hypotheses would be that the two groups (those assigned the nicotine lozenge and those assigned the placebo lozenge) are similar with regard to each of these background variables. The researchers would be hoping that they would not find a statistically significant difference between the groups on any of these variables so that a difference between the groups on the response variable could be attributed to the nicotine lozenge.

QUESTION 6

a.
Explanatory variable: positive or negative emotions (categorical - binary)
Response variable: whether or not subject caught a cold (categorical - binary)

b.
This was an observational study because the researchers observed and categorized the subjects' emotional states. They did not randomly assign subjects to be positive or negative thinkers.
Note: you might argue that this is an experiment because of the purposeful injection of the rhinovirus, but this was done for all groups. So, this study would not be considered a randomized comparative experiment from which we could draw a cause-and-effect conclusion.

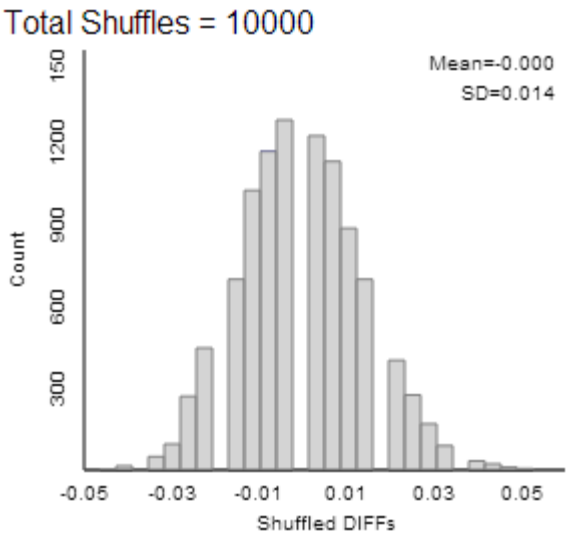
c.
No, because this was an observational study. There could be confounding variables, other factors that differ between those that are positive and negative thinkers, that explain the observed difference in likelihood of catching a cold. For example, maybe positive thinkers are more likely than negative thinkers to have a full breakfast every morning to start off their wonderful day.

QUESTION 7

a.
The observational unit is an eight-hour shift.

b.
Whether or not Gilbert worked the shift

c.



Statistic: $\hat{p}_{\text{DEATHGilbert}} - \hat{p}_{\text{DEATH.NoGilbert}} = 40/257 - 34/1384 = 0.131$

The observed statistic of 0.131 is nowhere near this graph. If we were to extend the *x*-axis, the statistic of 0.131 would appear in the extreme right of the graph. This would lead to a very small p-value, close to zero. The p-value would be one-sided because of the research question "investigating whether there was a higher risk of death when Gilbert was working," but there would be no data equal to the statistic of 0.131 or higher values. Such a small p-value would constitute very strong evidence against the null hypothesis, so we would reject the null hypothesis.

d.

We have sufficient evidence to say that the population proportion of all shifts whenb Gilbert is working with at least one death is higher than the population proprotion when Gilbert is not working, for eight hours ICU shifts at the Veteran's Administration hospital in Northampton, Massachusetts or similar to the ones in the study.

e.

No, despite the very small p-value, this is an **observational study** and we cannot draw cause-and-effect conclusions. Possible confounding variables could be the time of shift. If Gilbert was usually working overnight and overnight shifts tend to see more deaths than the other shifts.

f.

Standardized statistic (z): $z = \frac{(\hat{p}_1 - \hat{p}_2) - hyp.MEANnull}{SDnull} = (0.131 - 0)/0.014 = 9.29$

Because z = 9.29 is greater than 3 this is very strong evidence against the null hypothesis, we would reject the null hypothesis. The conclusion based on the z-statistic would be the same as the one based on the p-value.

Additional information to help you study: Since z = 9.29, we can interpret it and say that the observed difference in sample proportions of shifts with at least one death when Gilbert is working and not working of 0.131 is 9.29 standard deviations above zero (the hypothesized mean of the null distribution).

QUESTION 8

a.

	AZT	Placebo	Total
HIV	13	40	53

NO HIV	167	143	310
Total	180	183	363

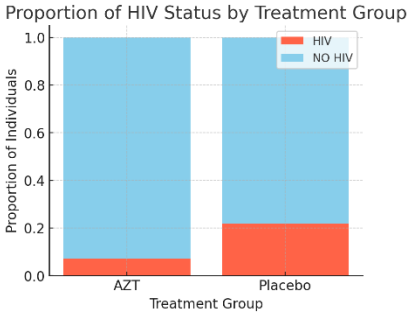
b.

$\hat{p}_{HIV_AZT} = 13/180 = 0.0722$

$\hat{p}_{HIV_PLACEBO} = 40/183 = 0.2186$

$\hat{p}_{HIV_PLACEBO} - \hat{p}_{HIV_AZT} = 0.2186 - 0.0722 = 0.1464$

This bar graph indicates that the sample proportion of babies who were HIV-infected given the placebo was larger than the sample proportion of babies who were HIV-infected given AZT. The difference in the percentages is 14.64 percentage points.



c.

The number of successes and failures in each group (each inner cell) should be at least 10.
AZT group: 13 HIV ("success") and 167 Not HIV ("failure")
PLACEBO group: 40 HIV babies ("success") and 143 Not HIV ("failure")
All four values are at least 10 so the validity conditions are met.

d.

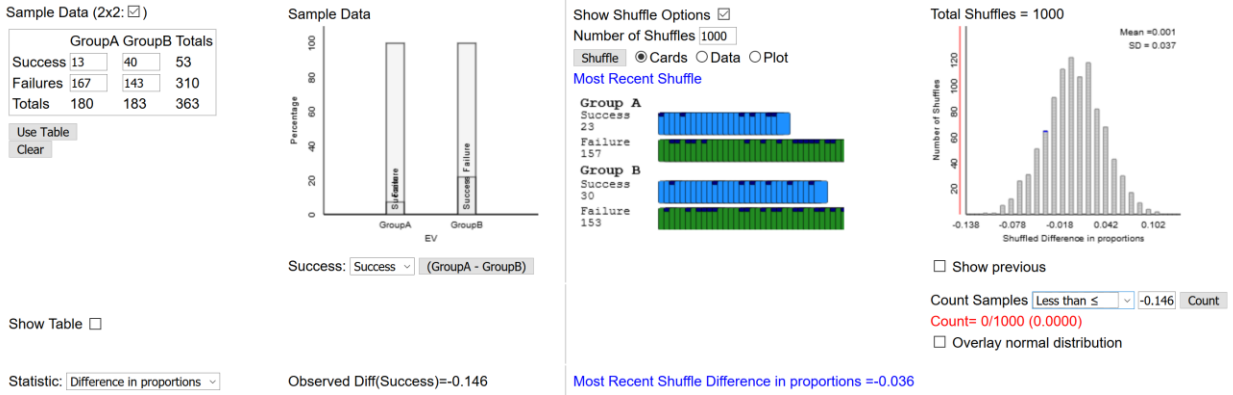
This study used random assignment of subjects to two treatment groups. We do not have information about the sampling, so we CANNOT assume that random sampling was used in this study.

e.

H_0: $\pi_{HIV_AZT} - \pi_{HIV_placebo.} = 0$

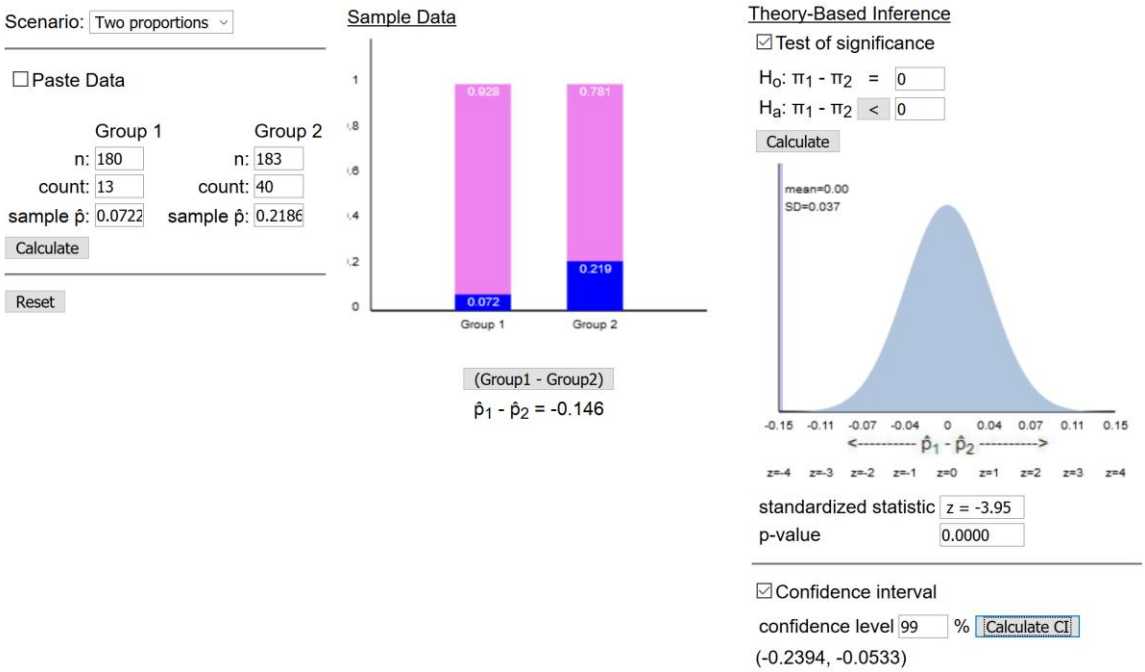
H_a: $\pi_{HIV_AZT} - \pi_{HIV_placebo} < 0$

We could enter the two-way table into the **Analyzing Two-way Tables** applet and find the p-value.



Statistic: $\hat{p}_{AZT} - \hat{p}_{placebo} = -0.146$
p-value = 0.000

Or, because the validity conditions are met, we could go straight to the Theory-Based applet to carry out a 'two-sample z-test'



Note: you wouldn't be asked to carry out these analyses but should be able to identify the appropriate analysis and interpret computer output.

f.

Type I error

g.
As the values in our interval are all negative and the order computed was AZT – PLACEBO, this implies that the AZT transmission rate is lower than the placebo transmission rate.

Interpretation: we are 99% confident that the population proportion of HIV-positive babies born to mothers who could potentially take AZT is between 0.0533 and 0.2394 smaller than the population proportion of HIV-positive babies born to mothers who could potentially take a placebo, for HIV-positive mothers (with babies) similar to the ones in this study.

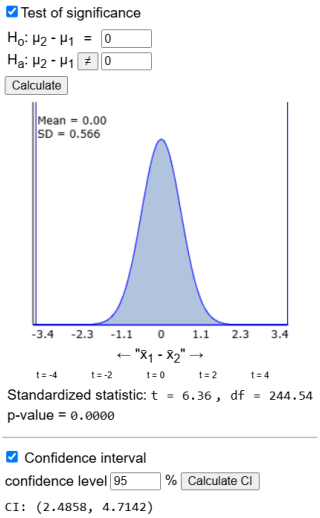
h.

Significance: Based on the very small p-value we can reject the null hypothesis and conclude that that the population proportion of HIV-positive babies born to mothers who could potentially take AZT is **smaller** than the proportion of HIV-positive babies born to mothers who could potentially take a placebo, for HIV-positive mothers (with babies) similar to the ones in this study. In other words, we have convincing evidence AZT is more effective than a placebo for reducing the probability of mother-to-infant transmission of AIDS.

Causation: because this was randomized experiment, we know that random assignment was used in the design of the study and possible confounding variables are most likely evenly split between groups. With the small p-value, we can conclude that AZT caused the observed difference in HIV transmission rates.
Generalization: we could generalize these results of this study to the population of HIV-positive mothers similar to the ones in this study.

QUESTION 9

a.
The t-statistic was 6.36 which means that the observed difference in sample means was 6.36 standard deviations above zero (the hypothesized mean of the null distribution). The theory-based p-value of 0.000 (and the t-statistic) gives us very strong evidence against the null hypothesis. We can reject the null hypothesis and say that the population mean number of friends differs between those with ages 65+ and those with ages 18–29, for subjects similar to the ones in this study.
We are 95% confident that the population mean number of friends reported by those who are 65+ is 2.486 to 4.714 friends higher than the mean for those who are ages 18-29, for subjects similar to the ones in this study.
If you want to interpret the p-value, you could say that the p-value of 0.000 is the probability of observing a difference in sample means like the one in the study or more extreme values on both tails, assuming that the null hypothesis is true.



b.
Theory-based approach

c.
1. If every person 65+ and every person with age between18-29 sampled had one more close friend than they originally reported, then the sample means for the two groups would increase by one. BUT the difference in sample means (statistic) would have remained the same. The sample standard deviations and sample sizes would also NOT change. The difference between the sample means would not change, so the standardized statistic value would not change, and thus the p-value would not change.

2. If both sample standard deviations were larger, the denominator of the standardized statistic would be larger (the SDnull would be larger), and thus the standardized statistic would be smaller, closer to the center of the distribution. This would make the p-value larger. OR you could also refer to the idea that with a larger variation in the null distribution (larger SDnull), we would have more data in the tail of the null distribution, and this would lead to a larger p-value.
Please note that you need to differentiate SAMPLE SD from SD_{NULL}.

3. If both sample sizes were larger, the denominator of the standardized statistic would be smaller (the SDnull would be smaller), and thus the standardized statistic would be larger (in absolute value), being more in the tail of the distribution. This would make the p-value smaller. OR you could also refer to the idea that with a smaller variation in the null distribution (smaller SDnull), we would have less data in the tail of the null distribution, and this would lead to a smaller p-value.

QUESTION 10

a.
One observational unit is one Adélie penguin.

b.
Explanatory variable: sex (female or male).
Response variable: body mass, measured in grams.

c.
$\bar{x}_M - \bar{x}_F = 4043.49 - 3368.84 = 674.65$ g

d.
H_0: $\mu_M - \mu_F = 0$
H_a: $\mu_M - \mu_F \neq 0$

e.
Let μ_M be the population mean body mass of male Adélie penguins and μF be the population mean body mass of female Adélie penguins.

f.
The t-statistic of 13.13 provides very strong evidence against the null hypothesis.
There is very strong evidence that the population mean body mass differs for male and female Adélie penguins in the Palmer Archipelago.

g.
We are 95% confident that the population mean body mass of male Adélie penguins is between 573.0g and 776.3g **higher** than the population mean body mass of female Adélie penguins, for Adélie penguins in the Palmer Archipelago.

h.
This was an observational study; sex was not randomly assigned, so confounding variables could be present and affecting the results of this study. Therefore, we cannot make a cause-and-effect conclusion.